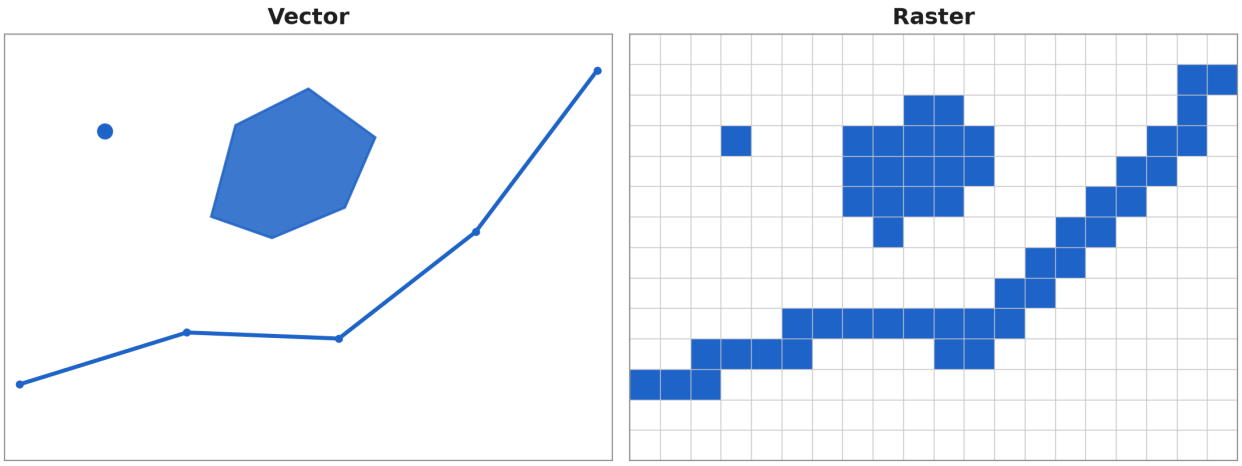
2.3 Rasters vs. vectors

A lot of spatial data can be represented as vector data as well as raster data. The choice between the two types depends on the **use case**.

A map with, for example, a boundary stone (point), a road (line) and a lake (polygon) would look like this as vector data (left) and as raster data (right):



	RASTER	VECTOR
Spatial definition	Based on a matrix of cells	Based on geometric objects
Spatial positioning	Each raster cell is implicitly given coordinates. This is based on the coordinates of the origin, the spatial resolution (real-world size of each cell) and the position of each cell in the matrix (row and column number).	Coordinates are explicitly stored for each point (or vertex).
File size	Usually larger files	Usually smaller files
Accuracy	Resolution-dependent, generalisation at cell level (loss of information)	No technical limits on accuracy
Suitable for	Continuous and discrete data	Discrete data (continuous data only in a simplified form, e.g. contour lines)
Attribute information	Raster cells can only contain numeric values. The number of values per cell is limited to one per *image band*.	Each object can have many attribute fields: text, numbers, dates…
Spatial analyses	Often faster because of the simple data structure	Often slower because of the more complex data structure